Data manipulation

Lecture 1

Elisa Alonso Herrero

M1 PPD - Fall 2026

Welcome to the course!

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About me

- PhD student at the Paris School of Economics
- I work primarily on:
  - Take-up of social programs
  - Optimal taxation
  - Intergenerational mobility
- I do empirical research, so I use Econometrics and (R) programming on a daily basis
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About the course

- Course originally designed by Louis Sirugue
- **Objective:** Learning R programming to carry out your empirical homeworks and research projects
- 4×2 hours:
  1. Data manipulation
  2. Data visualization
  3. R Markdown & LaTeX
  4. Econometrics and Tables in R

Let's delve into it!

1. Getting started

1.1. About R

- R is a **programming language** and free software environment for **statistical computing and graphics**

1. Getting started

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- The R language is widely (and increasingly) used in **academic and non-academic research** in fields like:
  - Economics
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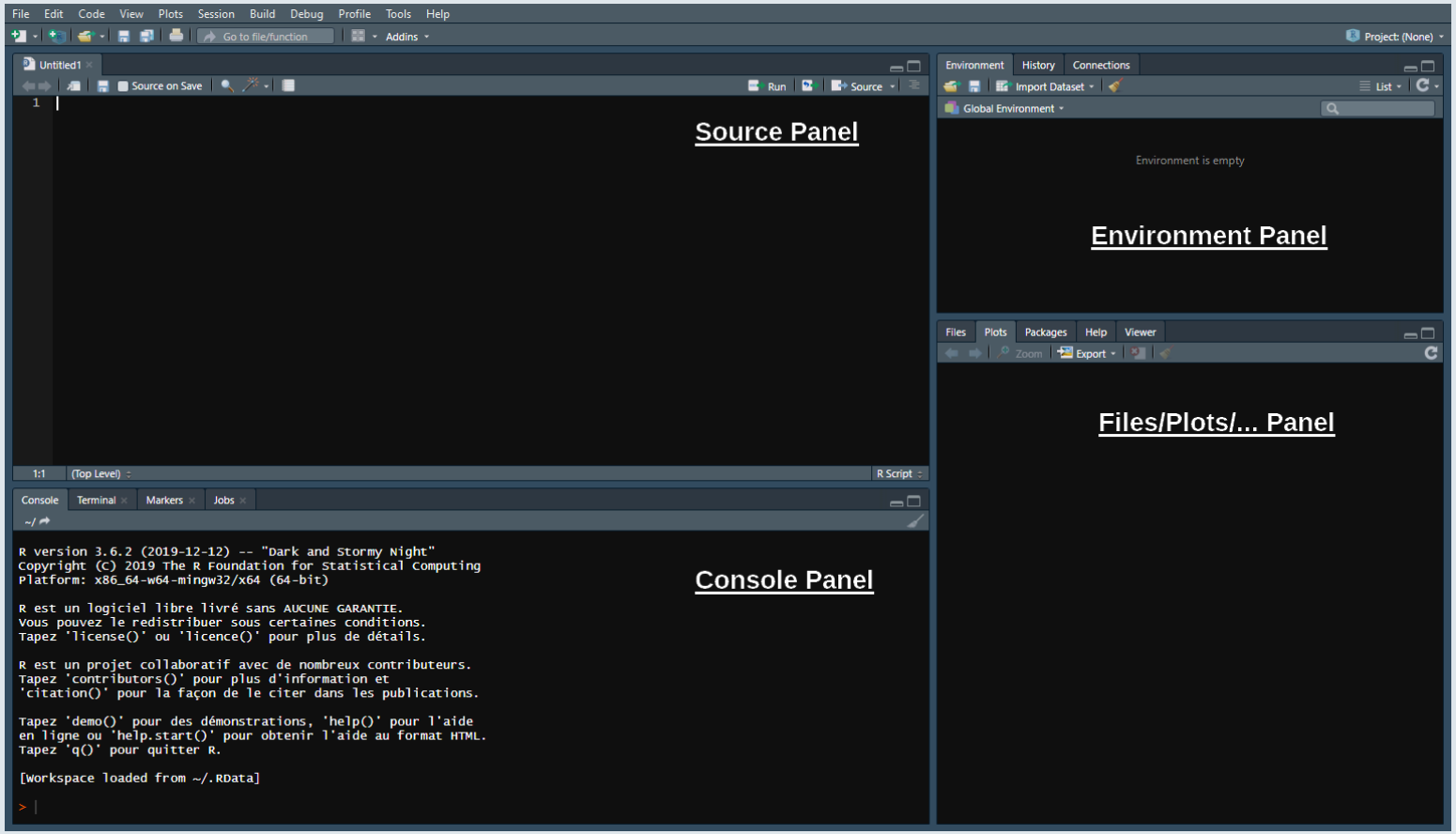
- The R language is widely (and increasingly) used in **academic and non-academic research** in fields like:
  - Economics
  - Statistics
  - Biostats
- Things you can do with R:
  - Reports
  - Nice plots
  - All the material of this course
  - Academic research
  - Win kaggle competitions
  - Interactive data visualization
  - Art

1. Getting started

1.2. The R Studio IDE



1. Getting started

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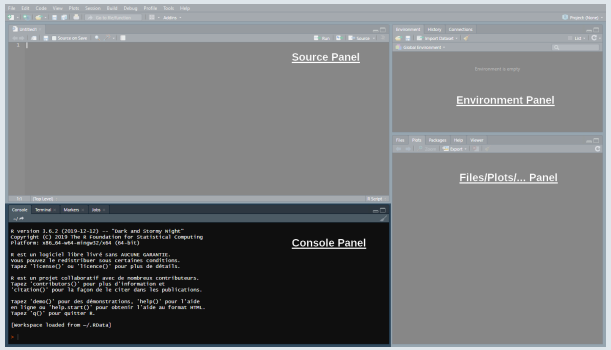

→ The Console panel

- This is where you **communicate with R**
  - You can write instructions after the **>**, **press enter** and R will **execute**
  - Try with **1+1:**

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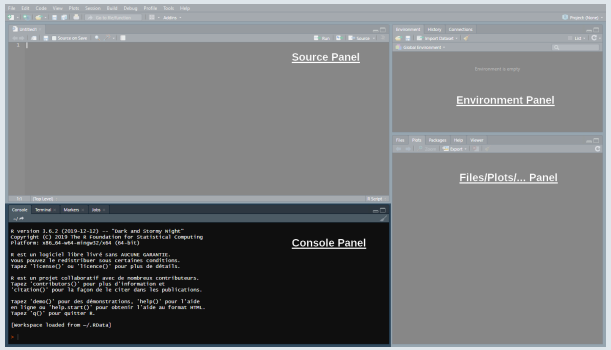

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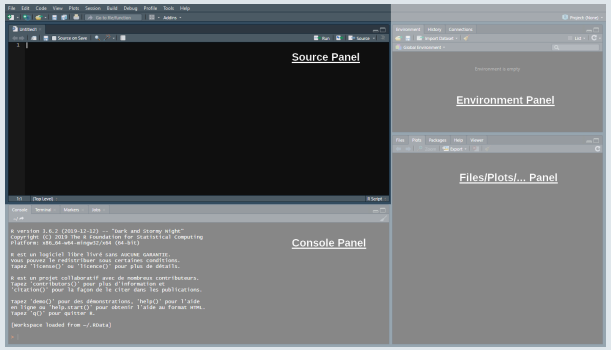

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- This is where you **write and save your code** (File > New File > R Script)
  - **Separate** different commands with a **line break**
  - The **#** symbol allows to **comment** your code
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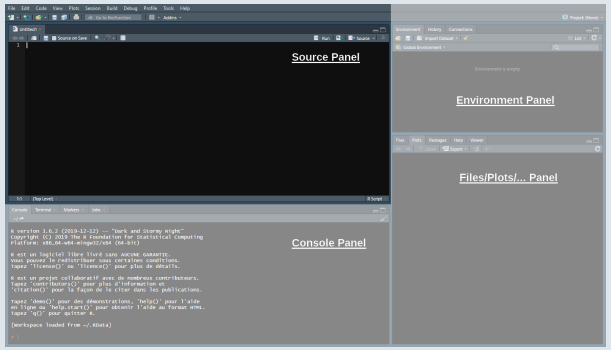

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```
1+1 # Do not put 2+2 on the same line, press enter to go to next line
2+2
```

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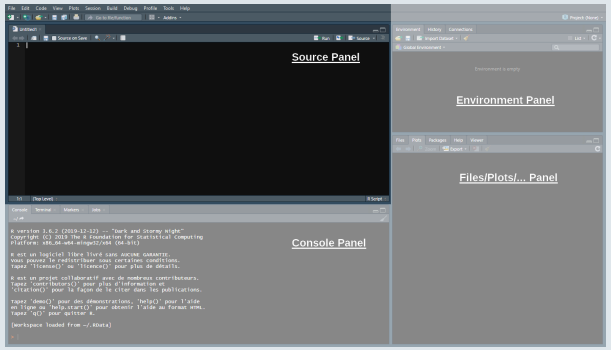

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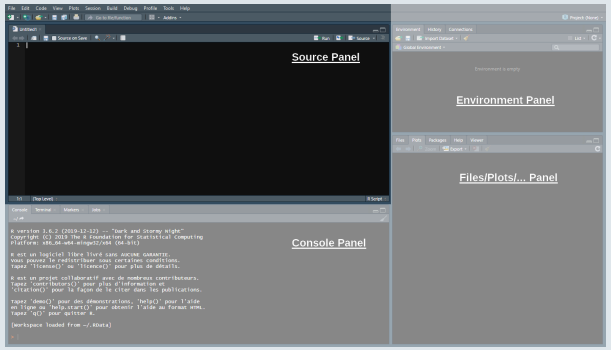

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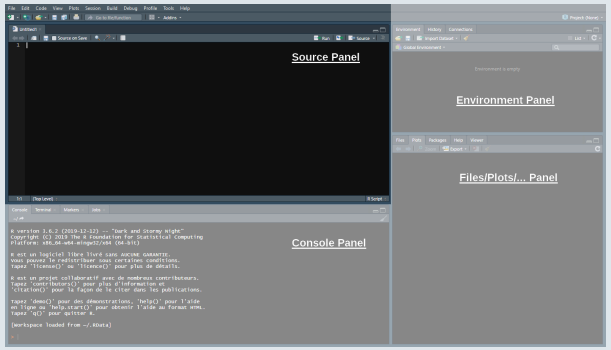

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- Check the console to see the output of your code

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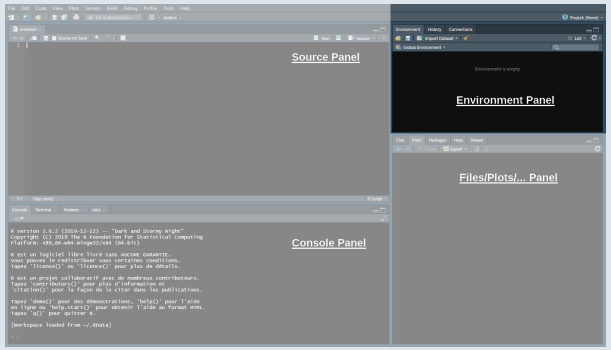

➜ The Environment panel

- Data analysis requires manipulating datasets, vectors, functions, etc.
  - These **elements are stored in the environment** panel

1. Getting started

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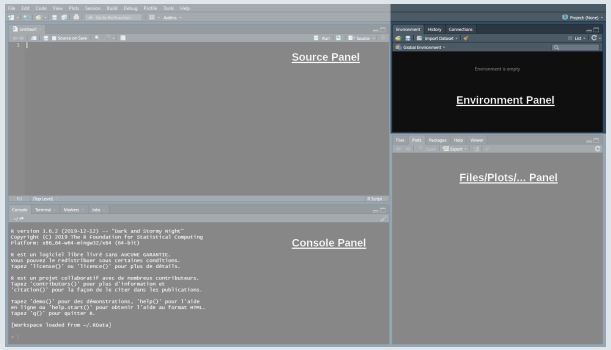

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- For instance we can assign a value to an object using `<-`

```
x <- 1
```

1. Getting started

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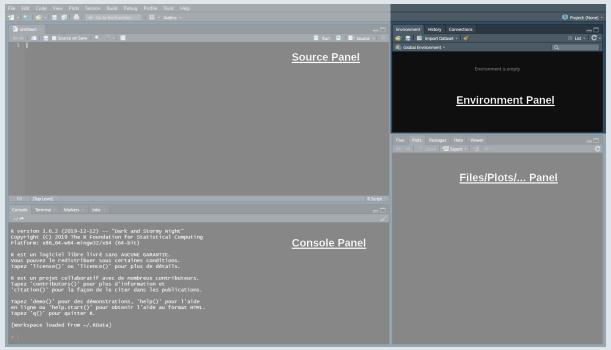

➜ The Environment panel

- Data analysis requires manipulating datasets, vectors, functions, etc.
  - These **elements are stored in the environment** panel
- For instance we can assign a value to an object using `<-`

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x <- 1
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➜ You now have an object called 'x' in your environment, which takes the value 1

1. Getting started

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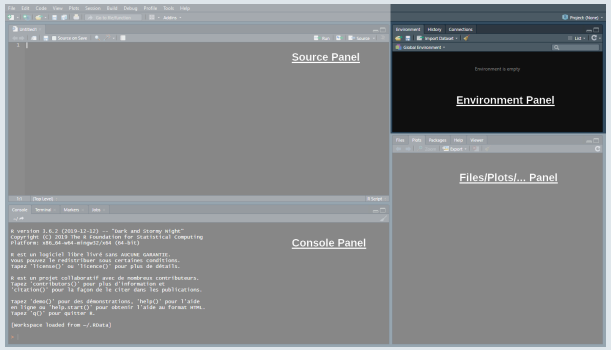

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- Now that the object `x` is stored in your environment, you can use it:

```
x + 1
```

```
## [1] 2
```

1. Getting started

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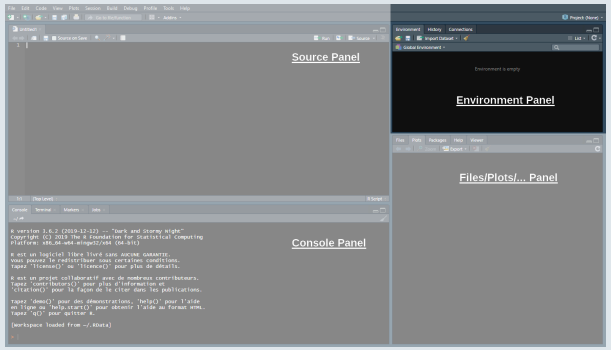

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- Now that the object `x` is stored in your environment, you can use it:

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```

```
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```

- You can also modify that object at any point:

```
x <- x + 1
x
```

```
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1. Getting started

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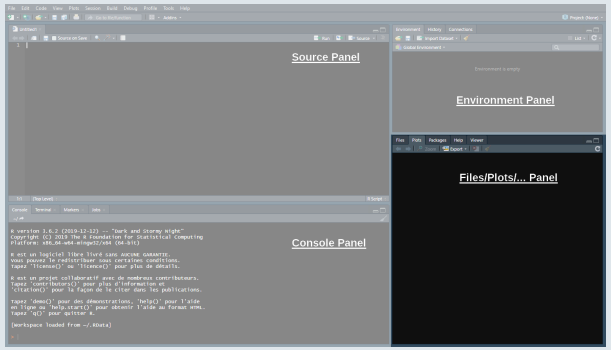

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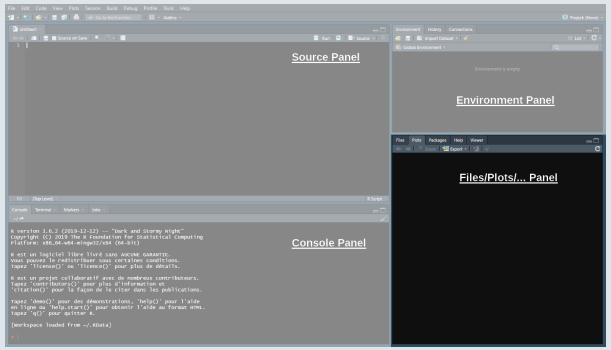
- In this panel we'll mainly be interested in the following 4 tabs

1. Getting started

1.2. The R Studio IDE



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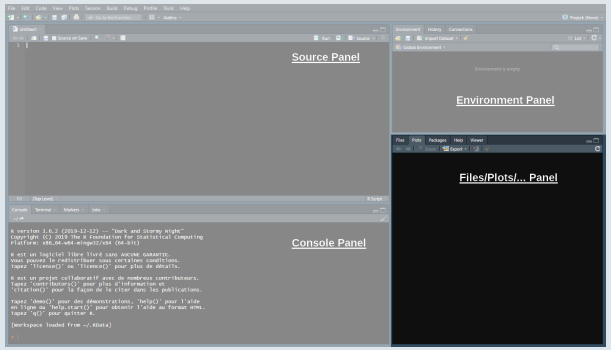

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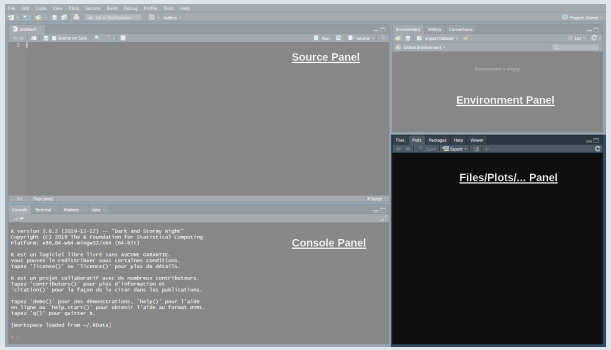

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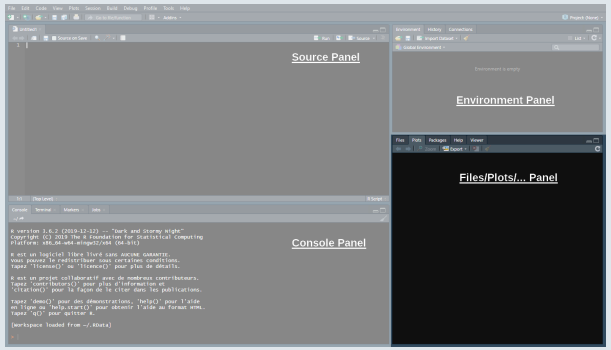

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  - **Files:** Shows your working directory
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  - **Packages:** A library of tools that we can load if needed
  - **Help:** Where to look for documentation on R functions

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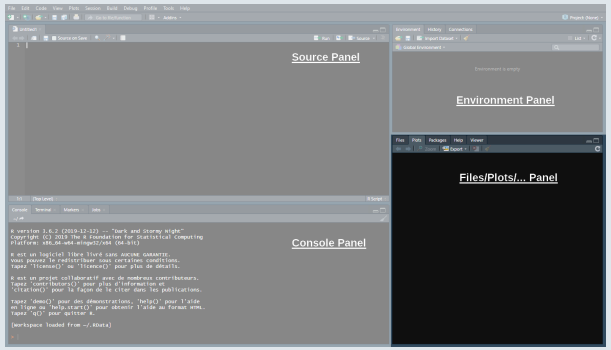

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- Enter `?getwd()` in the console to see what a **help file** looks like

1. Getting started

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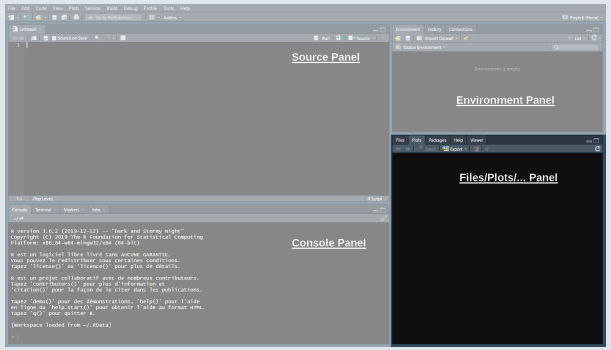
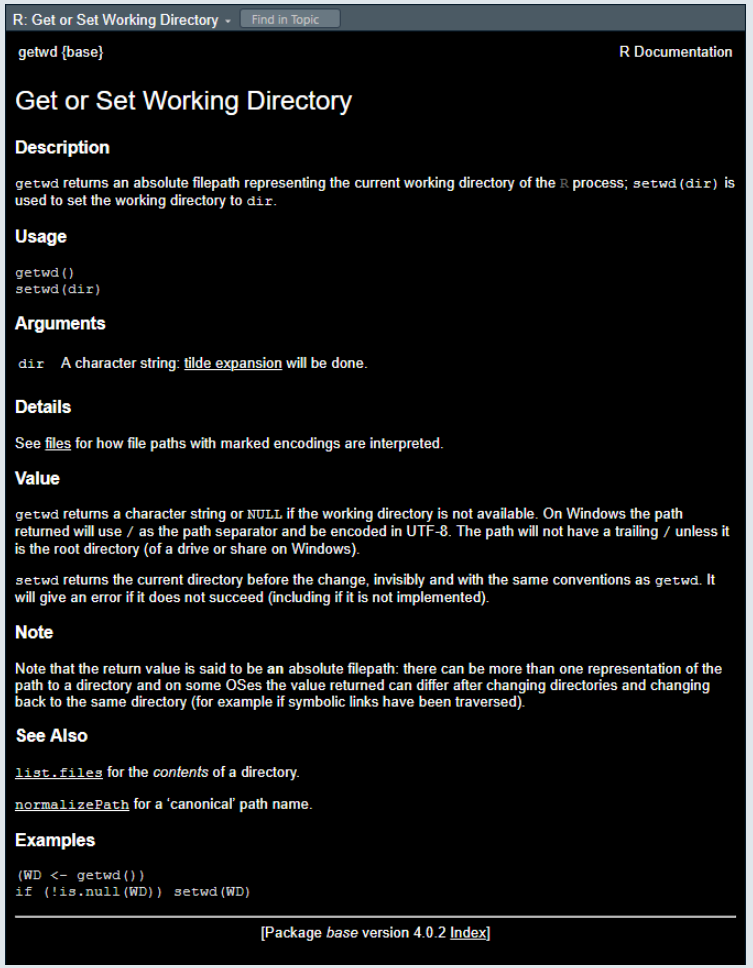


➜ The Files/Plots/... panel

- Enter `?getwd()` in the console to see what a **help file** looks like
  - It **describes** what the command does
  - It **explains** the different parameters of the command
  - It **gives examples** of how to use the command



R: Get or Set Working Directory — Find in Topic

getwd {base} — R Documentation

Get or Set Working Directory

Description

`getwd` returns an absolute filepath representing the current working directory of the `R` process; `setwd(dir)` is used to set the working directory to `dir`.

Usage

```
getwd()
setwd(dir)
```

Arguments

`dir` A character string: tilde expansion will be done.

Details

See files for how file paths with marked encodings are interpreted.

Value

`getwd` returns a character string or `NULL` if the working directory is not available. On Windows the path returned will use / as the path separator and be encoded in UTF-8. The path will not have a trailing / unless it is the root directory (of a drive or share on Windows).

`setwd` returns the current directory before the change, invisibly and with the same conventions as `getwd`. It will give an error if it does not succeed (including if it is not implemented).

Note

Note that the return value is said to be an absolute filepath: there can be more than one representation of the path to a directory and on some OSes the value returned can differ after changing directories and changing back to the same directory (for example if symbolic links have been traversed).

See Also

`list.files` for the *contents* of a directory.

`normalizePath` for a 'canonical' path name.

Examples

```
(WD <- getwd())
if (!is.null(WD)) setwd(WD)
```

[Package *base* version 4.0.2 Index]

1. Getting started

1.3. R projects

- What is an R project?
  - An R project is simply a **folder** tied to a **.Rproj** file
  - Opening the .Rproj file **starts a fresh R session** with the **working directory set to that folder**

1. Getting started

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- Why bother? This is **good practice** for anything that is not a one-off script
  - Your **scripts, data, and outputs** stay together in one place
  - Your code becomes **portable**: it will run the same way on a collaborator's computer, or on your computer in five years
  - You never need to use `setwd()` with an absolute path again

1. Getting started

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➜ *Create a project (File > New Project...) now for this course, and put today's script and data inside it*

Practice

1) Open a new R script (`Ctrl + Shift + N`) and write a code to create these objects:

Objects to create

Object name:	a	b	c
Assigned value:	2	4	5

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2) Run this code and create a new object named `result` that takes the value $\frac{b \times c}{a} + (b - a)^c$

Basic operations in R

Operation:	Addition	Subtraction	Multiplication	Division	Exponentiation	Parentheses
Symbol in R:	+	-	*	/	^	()

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02:59

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You've got 3 minutes!

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```
print(result)
```

```
## [1] 42
```

1. Getting started

1.4. Packages

- Base R is the core of the R programming language, but packages extend its capabilities (and make your life easier!):
  - Packages are **centralized** on the Comprehensive R Archive Network (CRAN)
  - To **download** and install a CRAN package you can simply use **install.packages()**

1. Getting started

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```
library(rio)
```

- You can also use a package's functions without fully loading the package by using the **double colon** operator (`package::function`)

1. Getting started

1.5. Import and eyeball data

- We now know how to **use R** as a calculator, but our goal is **to analyze data!**

1. Getting started

1.5. Import and eyeball data

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  ➜ *Take for instance the statistics from the last season of Ligue 1 available at fbref.com*

Scores & Fixtures 2021-2022 Ligue 1 Share & Export ▾ Glossary

All Rounds | Regular Season | French 1/2 Relegation/Promotion Play-offs

Wk	Day	Date	Time	Home	xG	Score	xG	Away	Attendance	Venue	Referee	Match Report	Notes
1	Fri	2021-08-06	21:00	Monaco	2.0	1–1	0.3	Nantes	7,500	Stade Louis II.	Antony Gautier	Match Report	
	Sat	2021-08-07	17:00	Lyon	1.4	1–1	0.8	Brest	29,018	Groupama Stadium	Mikael Lesage	Match Report	
			21:00	Troyes	0.8	1–2	1.2	**Paris S-G**	15,248	Stade de l'Aube	Amaury Delerue	Match Report	
	Sun	2021-08-08	13:00	Rennes	0.6	1–1	2.0	Lens	22,567	Roazhon Park	Bastien Dechepy	Match Report	
			15:00	Bordeaux	0.7	0–2	3.3	**Clermont Foot**	18,748	Stade Matmut-Atlantique	Florent Batta	Match Report	
			15:00	Strasbourg	0.4	0–2	0.9	**Angers**	23,250	Stade de la Meinau	Jeremy Stinat	Match Report	
			15:00	Nice	0.8	0–0	0.2	Reims	18,030	Stade de Nice	Johan Hamel	Match Report	
			15:00	Saint-Étienne	2.1	1–1	1.3	Lorient	20,461	Stade Geoffroy-Guichard	Thomas Léonard	Match Report	
			17:00	Metz	0.7	3–3	1.4	Lille	15,551	Stade Saint-Symphorien	Eric Wattellier	Match Report	
			20:45	Montpellier	0.5	2–3	2.0	**Marseille**	13,500	Stade de la Mosson	Jérémie Pignard	Match Report	
2	Fri	2021-08-13	21:00	**Lorient**	0.8	1–0	0.9	Monaco	12,149	Stade Yves Allainmat - Le Moustoir	Hakim Ben El Hadj Salem	Match Report	
	Sat	2021-08-14	17:00	Lille	1.0	0–4	2.2	**Nice**	30,144	Stade Pierre-Mauroy	Jérôme Brisard	Match Report	
			21:00	**Paris S-G**	2.5	4–2	0.6	Strasbourg	46,962	Parc des Princes	Willy Delajod	Match Report	
	Sun	2021-08-15	13:00	**Angers**	1.4	3–0	0.9	Lyon	6,154	Stade Raymond Kopa	Clément Turpin	Match Report	
			15:00	**Clermont Foot**	2.3	2–0	0.5	Troyes	11,005	Stade Gabriel Montpied	Romain Lissorgue	Match Report	
			15:00	Brest	1.5	1–1	0.9	Rennes	14,271	Stade Francis-Le Blé	Benoît Bastien	Match Report	

1. Getting started

1.5. Import and eyeball data

- You can **download** this dataset from Konosys or from the original course webpage
  - Note that the extension of the file is **.csv** (for ***C**omma **S**eparated **V**alues*)
  - Let's have a look at **first 5 lines** of the raw csv file

1. Getting started

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  - Let's have a look at **first 5 lines** of the raw csv file

```
Wk,Day,Date,Time,Home,xG,Score,xG,Away,Attendance,Venue,Referee,Match Report,Notes
1,Fri,2021-08-06,21:00,Monaco,2.0,1–1,0.3,Nantes,7500,Stade Louis II.,Antony Gautier,Match Report,
1,Sat,2021-08-07,17:00,Lyon,1.4,1–1,0.8,Brest,29018,Groupama Stadium,Mikael Lesage,Match Report,
1,Sat,2021-08-07,21:00,Troyes,0.8,1–2,1.2,Paris S-G,15248,Stade de l'Aube,Amaury Delerue,Match Report,
1,Sun,2021-08-08,13:00,Rennes,0.6,1–1,2.0,Lens,22567,Roazhon Park,Bastien Dechepy,Match Report,
```

1. Getting started

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  - Let's have a look at **first 5 lines** of the raw csv file

```
Wk,Day,Date,Time,Home,xG,Score,xG,Away,Attendance,Venue,Referee,Match Report,Notes
1,Fri,2021-08-06,21:00,Monaco,2.0,1–1,0.3,Nantes,7500,Stade Louis II.,Antony Gautier,Match Report,
1,Sat,2021-08-07,17:00,Lyon,1.4,1–1,0.8,Brest,29018,Groupama Stadium,Mikael Lesage,Match Report,
1,Sat,2021-08-07,21:00,Troyes,0.8,1–2,1.2,Paris S-G,15248,Stade de l'Aube,Amaury Delerue,Match Report,
1,Sun,2021-08-08,13:00,Rennes,0.6,1–1,2.0,Lens,22567,Roazhon Park,Bastien Dechepy,Match Report,
```

- The **.csv format** is very common and has a very **codified structure**
  - We can see that **each line** corresponds to **a row** (the first row generally contains column names)
  - And for each row the **values** of each column are **separated by commas**

1. Getting started

1.5. Import and eyeball data

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```
Wk,Day,Date,Time,Home,xG,Score,xG,Away,Attendance,Venue,Referee,Match Report,Notes
1,Fri,2021-08-06,21:00,Monaco,2.0,1–1,0.3,Nantes,7500,Stade Louis II.,Antony Gautier,Match Report,
1,Sat,2021-08-07,17:00,Lyon,1.4,1–1,0.8,Brest,29018,Groupama Stadium,Mikael Lesage,Match Report,
1,Sat,2021-08-07,21:00,Troyes,0.8,1–2,1.2,Paris S-G,15248,Stade de l'Aube,Amaury Delerue,Match Report,
1,Sun,2021-08-08,13:00,Rennes,0.6,1–1,2.0,Lens,22567,Roazhon Park,Bastien Dechepy,Match Report,
```

- The **.csv format** is very common and has a very **codified structure**
  - We can see that **each line** corresponds to **a row** (the first row generally contains column names)
  - And for each row the **values** of each column are **separated by commas**

➜ But how to get it in our R studio environment?

1. Getting started

1.5. Import and eyeball data

- There are many ways to **import** stuff in R, but I recommend the package `rio`
  - It takes the **file directory** as an **input**
  - And gives the **file content** as an **output**

```
function(input)
```

```
## [1] "output"
```

1. Getting started

1.5. Import and eyeball data

- There are many ways to **import** stuff in R, but I recommend the package `rio`
  - It takes the **file directory** as an **input**
  - And gives the **file content** as an **output**

```
function(input)
```

```
## [1] "output"
```

- The function `rio::import()` chooses the right reader **based on the file extension** (.csv, .xlsx, .dta, .sav, ...)
- For **.csv** files, it uses **data.table::fread()** under the hood, which is **much faster** than base R's reader

- Let's use it to import the Ligue 1 data

```
fb <- rio::import("C:/User/Documents/R class/lecture1/ligue1.csv")
```

➜ This works, but an absolute path is a bad idea... there's a better way!

1. Getting started

1.5. Import and eyeball data

- An **absolute path** like `"C:/User/Documents/class/ligue1.csv"` only works **on your computer**
  - It **breaks** as soon as you share your script, or move your files around
  - Depending on your operating system, it can also break if you use backslashes (\) instead of forward slashes (/)

1. Getting started

1.5. Import and eyeball data

- An **absolute path** like `"C:/User/Documents/class/ligue1.csv"` only works **on your computer**
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  - Depending on your operating system, it can also break if you use backslashes (\) instead of forward slashes (/)

- Since we are working inside an **R project**, we can use **relative paths** instead
  - They are defined **from the project's root folder**, not from your computer's root

1. Getting started

1.5. Import and eyeball data

- An **absolute path** like `"C:/User/Documents/class/ligue1.csv"` only works **on your computer**
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- Since we are working inside an **R project**, we can use **relative paths** instead
  - They are defined **from the project's root folder**, not from your computer's root

- The `here` package makes this effortless with the **here()** function
  - It automatically finds your **project's root**, wherever your R script is opened from
  - You just describe the path to your file **from there**

1. Getting started

1.5. Import and eyeball data

- An **absolute path** like `"C:/User/Documents/class/ligue1.csv"` only works **on your computer**
  - It **breaks** as soon as you share your script, or move your files around
  - Depending on your operating system, it can also break if you use backslashes (\) instead of forward slashes (/)

- Since we are working inside an **R project**, we can use **relative paths** instead
  - They are defined **from the project's root folder**, not from your computer's root

- The `here` package makes this effortless with the **here()** function
  - It automatically finds your **project's root**, wherever your R script is opened from
  - You just describe the path to your file **from there**

```
fb <- rio::import(here::here("lecture1", "ligue1.csv"))
```

➜ *Let's **inspect** this new object to check that it worked*

1. Getting started

1.5. Import and eyeball data

- The first thing we can do is to use `head()` to print the **top rows**

```
head(fb, 4)
```

1. Getting started

1.5. Import and eyeball data

- The first thing we can do is to use `head()` to print the **top rows**

```
head(fb, 4)
```

```
##   Wk Day       Date  Time   Home  xG Score xG.1      Away Attendance
## 1  1 Fri 2021-08-06 21:00 Monaco 2.0   1-1  0.3    Nantes       7500
## 2  1 Sat 2021-08-07 17:00   Lyon 1.4   1-1  0.8     Brest      29018
## 3  1 Sat 2021-08-07 21:00 Troyes 0.8   1-2  1.2 Paris S-G      15248
## 4  1 Sun 2021-08-08 13:00 Rennes 0.6   1-1  2.0      Lens      22567
##              Venue         Referee Match.Report Notes
## 1  Stade Louis II.  Antony Gautier Match Report    NA
## 2 Groupama Stadium   Mikael Lesage Match Report    NA
## 3  Stade de l'Aube  Amaury Delerue Match Report    NA
## 4     Roazhon Park Bastien Dechepy Match Report    NA
```

1. Getting started

1.5. Import and eyeball data

- The first thing we can do is to use `head()` to print the **top rows**

```
head(fb, 4)
```

```
##   Wk Day       Date  Time   Home  xG Score xG.1      Away Attendance
## 1  1 Fri 2021-08-06 21:00 Monaco 2.0   1-1  0.3    Nantes       7500
## 2  1 Sat 2021-08-07 17:00   Lyon 1.4   1-1  0.8     Brest      29018
## 3  1 Sat 2021-08-07 21:00 Troyes 0.8   1-2  1.2 Paris S-G      15248
## 4  1 Sun 2021-08-08 13:00 Rennes 0.6   1-1  2.0      Lens      22567
##              Venue         Referee Match.Report Notes
## 1  Stade Louis II.  Antony Gautier Match Report    NA
## 2 Groupama Stadium   Mikael Lesage Match Report    NA
## 3  Stade de l'Aube  Amaury Delerue Match Report    NA
## 4     Roazhon Park Bastien Dechepy Match Report    NA
```

- `tail()` would print the **bottom rows**
- We can also run **`View(fb)`** *(a new tab will pop-up in your Source panel)*

1. Getting started

1.5. Import and eyeball data

	Wk	Day	Date	Time	Home	xG	Score	xG.1	Away	Attendance	Venue	Referee	Match.Report	Notes
1	1	Fri	2021-08-06	21:00	Monaco	2.0	1-1	0.3	Nantes	7500	Stade Louis II.	Antony Gautier	Match Report	*NA*
2	1	Sat	2021-08-07	17:00	Lyon	1.4	1-1	0.8	Brest	29018	Groupama Stadium	Mikael Lesage	Match Report	*NA*
3	1	Sat	2021-08-07	21:00	Troyes	0.8	1-2	1.2	Paris S-G	15248	Stade de l'Aube	Amaury Delerue	Match Report	*NA*
4	1	Sun	2021-08-08	13:00	Rennes	0.6	1-1	2.0	Lens	22567	Roazhon Park	Bastien Dechepy	Match Report	*NA*
5	1	Sun	2021-08-08	15:00	Bordeaux	0.7	0-2	3.3	Clermont Foot	18748	Stade Matmut-Atlantique	Florent Batta	Match Report	*NA*
6	1	Sun	2021-08-08	15:00	Strasbourg	0.4	0-2	0.9	Angers	23250	Stade de la Meinau	Jeremy Stinat	Match Report	*NA*
7	1	Sun	2021-08-08	15:00	Nice	0.8	0-0	0.2	Reims	18030	Stade de Nice	Johan Hamel	Match Report	*NA*
8	1	Sun	2021-08-08	15:00	Saint-Étienne	2.1	1-1	1.3	Lorient	20461	Stade Geoffroy-Guichard	Thomas Léonard	Match Report	*NA*
9	1	Sun	2021-08-08	17:00	Metz	0.7	3-3	1.4	Lille	15551	Stade Saint-Symphorien	Eric Wattellier	Match Report	*NA*
10	1	Sun	2021-08-08	20:45	Montpellier	0.5	2-3	2.0	Marseille	13500	Stade de la Mosson	Jérémie Pignard	Match Report	*NA*
11	2	Fri	2021-08-13	21:00	Lorient	0.8	1-0	0.9	Monaco	12149	Stade Yves Allainmat - Le Moustoir	Hakim Ben El Hadj Salem	Match Report	*NA*
12	2	Sat	2021-08-14	17:00	Lille	1.0	0-4	2.2	Nice	30144	Stade Pierre-Mauroy	Jérôme Brisard	Match Report	*NA*
13	2	Sat	2021-08-14	21:00	Paris S-G	2.5	4-2	0.6	Strasbourg	46962	Parc des Princes	Willy Delajod	Match Report	*NA*
14	2	Sun	2021-08-15	13:00	Angers	1.4	3-0	0.9	Lyon	6154	Stade Raymond Kopa	Clément Turpin	Match Report	*NA*
15	2	Sun	2021-08-15	15:00	Clermont Foot	2.3	2-0	0.5	Troyes	11005	Stade Gabriel Montpied	Romain Lissorgue	Match Report	*NA*
16	2	Sun	2021-08-15	15:00	Brest	1.5	1-1	0.9	Rennes	14271	Stade Francis-Le Blé	Benoît Bastien	Match Report	*NA*
17	2	Sun	2021-08-15	15:00	Nantes	1.4	2-0	1.1	Metz	12054	Stade de la Beaujoire - Louis Fonteneau	Johan Hamel	Match Report	*NA*

Seems like it worked, but always check your data!

Overview

1. Getting started ✓
- 1.1. About R
- 1.2. The R Studio IDE
- 1.3. R projects
- 1.4. Packages
- 1.5. Import and eyeball data

2. Anatomy of a data.frame
- 2.1. Data structure
- 2.2. Classes
- 2.3. Vectors
- 2.4. Subsetting

3. The dplyr package
- 3.1. Basic functions
- 3.2. group_by() and summarise()

4. A few words on learning R
- 4.1. When it doesn't work the way you want
- 4.2. Where to find help
- 4.3. When it doesn't work at all

5. Wrap up!

2. Anatomy of a `data.frame`

2.1. Data structure

- Now that we imported the data properly, we can check out its **`str()`ucture** in more details

```
str(fb)
```

2. Anatomy of a `data.frame`

2.1. Data structure

- *Don't be scared of the output!*

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
##  $ Wk          : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Day         : chr  "Fri" "Sat" "Sat" "Sun" ...
##  $ Date        : IDate, format: "2021-08-06" "2021-08-07" ...
##  $ Time        : chr  "21:00" "17:00" "21:00" "13:00" ...
##  $ Home        : chr  "Monaco" "Lyon" "Troyes" "Rennes" ...
##  $ xG          : num  2 1.4 0.8 0.6 0.7 0.4 0.8 2.1 0.7 0.5 ...
##  $ Score       : chr  "1-1" "1-1" "1-2" "1-1" ...
##  $ xG.1        : num  0.3 0.8 1.2 2 3.3 0.9 0.2 1.3 1.4 2 ...
##  $ Away        : chr  "Nantes" "Brest" "Paris S-G" "Lens" ...
##  $ Attendance  : int  7500 29018 15248 22567 18748 23250 18030 20461 15551 13500 ...
##  $ Venue       : chr  "Stade Louis II." "Groupama Stadium" "Stade de l'Aube" "Roazhon Park" ...
##  $ Referee     : chr  "Antony Gautier" "Mikael Lesage" "Amaury Delerue" "Bastien Dechepy" ...
##  $ Match.Report: chr  "Match Report" "Match Report" "Match Report" "Match Report" ...
##  $ Notes       : logi  NA NA NA NA NA NA ...
```

2. Anatomy of a `data.frame`

2.1. Data structure

- `str()` says that `fb` is a `data.frame`, and gives its numbers of **observations** (rows) and **variables** (columns)

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
```

2. Anatomy of a `data.frame`

2.1. Data structure

- It also gives the **variables names**

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
##  $ Wk
##  $ Day
##  $ Date
##  $ Time
##  $ Home
##  $ xG
##  $ Score
##  $ xG.1
##  $ Away
##  $ Attendance
##  $ Venue
##  $ Referee
##  $ Match.Report
##  $ Notes
```

2. Anatomy of a `data.frame`

2.1. Data structure

- The **first values** of each variable

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
##  $ Wk          :      1 1 1 1 1 1 1 1 1 1 ...
##  $ Day         :      "Fri" "Sat" "Sat" "Sun" ...
##  $ Date        :      "2021-08-06" "2021-08-07" "2021-08-07" "2021-08-08" ...
##  $ Time        :      "21:00" "17:00" "21:00" "13:00" ...
##  $ Home        :      "Monaco" "Lyon" "Troyes" "Rennes" ...
##  $ xG          :      2 1.4 0.8 0.6 0.7 0.4 0.8 2.1 0.7 0.5 ...
##  $ Score       :      "1-1" "1-1" "1-2" "1-1" ...
##  $ xG.1        :      0.3 0.8 1.2 2 3.3 0.9 0.2 1.3 1.4 2 ...
##  $ Away        :      "Nantes" "Brest" "Paris S-G" "Lens" ...
##  $ Attendance  :      7500 29018 15248 22567 18748 23250 18030 20461 15551 13500 ...
##  $ Venue       :      "Stade Louis II." "Groupama Stadium" "Stade de l'Aube" "Roazhon Park" ...
##  $ Referee     :      "Antony Gautier" "Mikael Lesage" "Amaury Delerue" "Bastien Dechepy" ...
##  $ Match.Report:      "Match Report" "Match Report" "Match Report" "Match Report" ...
##  $ Notes       :       NA NA NA NA NA NA ...
```

2. Anatomy of a `data.frame`

2.1. Data structure

- As well as the **class** of each variable

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
##  $ Wk          : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Day         : chr  "Fri" "Sat" "Sat" "Sun" ...
##  $ Date        : chr  "2021-08-06" "2021-08-07" "2021-08-07" "2021-08-08" ...
##  $ Time        : chr  "21:00" "17:00" "21:00" "13:00" ...
##  $ Home        : chr  "Monaco" "Lyon" "Troyes" "Rennes" ...
##  $ xG          : num  2 1.4 0.8 0.6 0.7 0.4 0.8 2.1 0.7 0.5 ...
##  $ Score       : chr  "1-1" "1-1" "1-2" "1-1" ...
##  $ xG.1        : num  0.3 0.8 1.2 2 3.3 0.9 0.2 1.3 1.4 2 ...
##  $ Away        : chr  "Nantes" "Brest" "Paris S-G" "Lens" ...
##  $ Attendance  : int  7500 29018 15248 22567 18748 23250 18030 20461 15551 13500 ...
##  $ Venue       : chr  "Stade Louis II." "Groupama Stadium" "Stade de l'Aube" "Roazhon Park" ...
##  $ Referee     : chr  "Antony Gautier" "Mikael Lesage" "Amaury Delerue" "Bastien Dechepy" ...
##  $ Match.Report: chr  "Match Report" "Match Report" "Match Report" "Match Report" ...
##  $ Notes       : logi  NA NA NA NA NA NA ...
```

2. Anatomy of a `data.frame`

2.1. Data structure

- But what does the **class** correspond to?

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
##  $ Wk          : int  ?
##  $ Day         : chr  ?
##  $ Date        : chr  ?
##  $ Time        : chr  ?
##  $ Home        : chr  ?
##  $ xG          : num  ?
##  $ Score       : chr  ?
##  $ xG.1        : num  ?
##  $ Away        : chr  ?
##  $ Attendance  : int  ?
##  $ Venue       : chr  ?
##  $ Referee     : chr  ?
##  $ Match.Report: chr  ?
##  $ Notes       : logi  ?
```

2. Anatomy of a `data.frame`

2.2. Classes

Numeric	Character	Logical

2. Anatomy of a `data.frame`

2.2. Classes

Numeric **Character** **Logical**

These are simply numbers:

```
class(3)
```

```
## [1] "numeric"
```

```
class(-1.6180339)
```

```
## [1] "numeric"
```

Numeric variable classes include:

- **int** for round numbers
- **dbl** for 2-decimal numbers

2. Anatomy of a `data.frame`

2.2. Classes

Numeric

These are simply numbers:

```
class(3)
```

```
## [1] "numeric"
```

```
class(-1.6180339)
```

```
## [1] "numeric"
```

Numeric variable classes include:

- **int** for round numbers
- **dbl** for 2-decimal numbers

Character

They must be surrounded by `"`:

```
class("Roazhon Park")
```

```
## [1] "character"
```

```
class("35")
```

```
## [1] "character"
```

We also call these values:

- Character strings
- Or just strings

Logical

2. Anatomy of a `data.frame`

2.2. Classes

Numeric

These are simply numbers:

```
class(3)
```

```
## [1] "numeric"
```

```
class(-1.6180339)
```

```
## [1] "numeric"
```

Numeric variable classes include:

- **int** for round numbers
- **dbl** for 2-decimal numbers

Character

They must be surrounded by `"`:

```
class("Roazhon Park")
```

```
## [1] "character"
```

```
class("35")
```

```
## [1] "character"
```

We also call these values:

- Character strings
- Or just strings

Logical

Something either `TRUE` of `FALSE`:

```
3 >= 4
```

```
## [1] FALSE
```

```
class(T)
```

```
## [1] "logical"
```

Most common logical operators:

- == > < <= >=
- & (and) | (or) ! (opposite)

2. Anatomy of a `data.frame`

2.2. Classes

Guess the output!

```
as.numeric("2022")
```

2. Anatomy of a `data.frame`

2.2. Classes

Guess the output!

```
as.numeric("2022")
```

```
## [1] 2022
```

2. Anatomy of a `data.frame`

2.2. Classes

Guess the output!

```
as.numeric("2022")
```

```
## [1] 2022
```

What about this one?

```
as.character(2022-2023)
```

2. Anatomy of a `data.frame`

2.2. Classes

Guess the output!

```
as.numeric("2022")
```

```
## [1] 2022
```

What about this one?

```
as.character(2022-2023)
```

```
## [1] "-1"
```

2. Anatomy of a `data.frame`

2.2. Classes

Guess the output!

```
as.numeric("2022")
```

```
## [1] 2022
```

What about this one?

```
as.character(2022-2023)
```

```
## [1] "-1"
```

A final one:

```
as.character(2022>2023)
```

2. Anatomy of a `data.frame`

2.2. Classes

Guess the output!

```
as.numeric("2022")
```

```
## [1] 2022
```

What about this one?

```
as.character(2022-2023)
```

```
## [1] "-1"
```

A final one:

```
as.character(2022>2023)
```

```
## [1] "FALSE"
```

2. Anatomy of a `data.frame`

2.2. Classes

- To know everything:

Class conversion table:

	numeric	**character**	**logical**
as.numeric()	No effect	Converts strings of numbers into numeric values Returns NA if characters in the string	Returns 1 if TRUE Returns 0 if FALSE
as.character()	Converts numeric values into strings of numbers	No effect	Returns "TRUE" if TRUE Returns "FALSE" if FALSE
as.logical()	Returns TRUE if != 0 Returns FALSE if 0	Returns TRUE if "T" or"TRUE" Returns FALSE if "F" or "FALSE" Returns NA otherwise	No effect

NA stands for *'Not Available'*, it corresponds to a **missing value**

2. Anatomy of a `data.frame`

2.2. Classes

- Great! But there is one last mystery...

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
##  $ Wk          : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Day         : chr  "Fri" "Sat" "Sat" "Sun" ...
##  $ Date        : chr  "2021-08-06" "2021-08-07" "2021-08-07" "2021-08-08" ...
##  $ Time        : chr  "21:00" "17:00" "21:00" "13:00" ...
##  $ Home        : chr  "Monaco" "Lyon" "Troyes" "Rennes" ...
##  $ xG          : num  2 1.4 0.8 0.6 0.7 0.4 0.8 2.1 0.7 0.5 ...
##  $ Score       : chr  "1-1" "1-1" "1-2" "1-1" ...
##  $ xG.1        : num  0.3 0.8 1.2 2 3.3 0.9 0.2 1.3 1.4 2 ...
##  $ Away        : chr  "Nantes" "Brest" "Paris S-G" "Lens" ...
##  $ Attendance  : int  7500 29018 15248 22567 18748 23250 18030 20461 15551 13500 ...
##  $ Venue       : chr  "Stade Louis II." "Groupama Stadium" "Stade de l'Aube" "Roazhon Park" ...
##  $ Referee     : chr  "Antony Gautier" "Mikael Lesage" "Amaury Delerue" "Bastien Dechepy" ...
##  $ Match.Report: chr  "Match Report" "Match Report" "Match Report" "Match Report" ...
##  $ Notes       : logi  NA NA NA NA NA NA ...
```

2. Anatomy of a `data.frame`

2.2. Classes

- Are these dollar signs here for a reason?

```
str(fb)
```

```
## 'data.frame':    380 obs. of  14 variables:
##  $ Wk
##  $ Day
##  $ Date
##  $ Time
##  $ Home
##  $ xG
##  $ Score
##  $ xG.1
##  $ Away
##  $ Attendance
##  $ Venue
##  $ Referee
##  $ Match.Report
##  $ Notes
```

2. Anatomy of a `data.frame`

2.3. Vectors

- It's actually just a reference to the fact that `$` allows to **extract a variable** from a dataset

2. Anatomy of a `data.frame`

2.3. Vectors

- It's actually just a reference to the fact that **`$`** allows to **extract a variable** from a dataset

```
fb$Wk
```

```
##   [1]  1  1  1  1  1  1  1  1  1  1  2  2  2  2  2  2  2  2  2  2  3  3  3  3  3  3  3  3  3
##  [30]  4  4  4  4  4  4  4  4  4  4  5  5  5  5  5  5  5  5  5  5  6  6  6  6  6  6  6  6  6
##  [59]  6  7  7  7  7  7  7  7  7  7  7  8  8  8  8  8  8  8  8  8  8  8  9  9  9  9  9  9  9  9
##  [88]  9  9 10 10 10 10 10 10 10 10 10 10 11 11 11 11 11 11 11 11 11 11 11  3 12 12 12 12 12 12
## [117] 12 12 12 12 13 13 13 13 13 13 13 13 13 13 14 14 14 14 14 14 14 14 14 14 15 15 15 15 15 15
## [146] 15 15 15 15 16 16 16 16 16 16 16 16 16 16 16 17 17 17 17 17 17 17 17 17 17 17 18 18 18 18 18
## [175] 18 18 18 18 18 19 19 19 19 19 19 19 19 19 19 20 20 20 20 20 20 20 21 21 21 21 21 21 21 21
## [204] 21 21 19 20 20 22 22 22 22 22 22 22 22 22 22 22 20 14 23 23 23 23 23 23 23 23 23 23 24 24
## [233] 24 24 24 24 24 24 24 24 25 25 25 25 25 25 25 25 25 25 25 26 26 26 26 26 26 26 26 26 26 27
## [262] 27 27 27 27 27 27 27 27 27 28 28 28 28 28 28 28 28 28 28 29 29 29 29 29 29 29 29 29 29 29
## [291] 30 30 30 30 30 30 30 30 30 30 31 31 31 31 31 31 31 31 31 31 32 32 32 32 32 32 32 32 32 32
## [320] 32 33 33 33 33 33 33 33 33 33 33 34 34 34 34 34 34 34 34 34 34 35 35 35 35 35 35 35 35 35
## [349] 35 35 36 36 36 36 36 36 36 36 36 36 37 37 37 37 37 37 37 37 37 37 38 38 38 38 38 38 38 38
## [378] 38 38 38
```

2. Anatomy of a `data.frame`

2.3. Vectors

- We call these objects **vectors**
  - Vectors are basically **sequences of values that have the same class**
  - R won't let you create a vector containing elements of different classes

2. Anatomy of a `data.frame`

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- We make our own vectors using the **`c()`oncatenate** function

2. Anatomy of a `data.frame`

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- We make our own vectors using the **`c()`oncatenate** function

```
c("Hello world", 35, FALSE)
```

```
## [1] "Hello world" "35"          "FALSE"
```

2. Anatomy of a `data.frame`

2.3. Vectors

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```
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- The fact that vectors are homogeneous in class allows that **operations apply to all their elements**

2. Anatomy of a `data.frame`

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## [1] "Hello world" "35"          "FALSE"
```

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```
c(1, 2, 3) / 3
```

```
## [1] 0.3333333 0.6666667 1.0000000
```

2. Anatomy of a `data.frame`

2.3. Vectors

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```
c("Hello world", 35, FALSE)
```

```
## [1] "Hello world" "35"          "FALSE"
```

- The fact that vectors are homogeneous in class allows that **operations apply to all their elements**

```
c(1, 2, 3) / 3
```

```
## [1] 0.3333333 0.6666667 1.0000000
```

```
3 / c(1, 2, 3)
```

```
## [1] 3.0 1.5 1.0
```

2. Anatomy of a `data.frame`

2.4. Subsetting

- But **$** is not the only way to **extract** a variable from a dataset
  - You can also make use of the **[ ]** subsetting operator

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$$\text{data}[\text{row},\ \ \text{columns}]$$

2. Anatomy of a `data.frame`

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- But **$** is not the only way to **extract** a variable from a dataset
  - You can also make use of the **[ ]** subsetting operator

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- Inside the **brackets**, indicate what you want to **keep using:**
  - **Indices:** e.g., the third column has index 3
  - **Logical:** A vector of TRUE and FALSE
  - **Names:** They must be in quotation marks

2. Anatomy of a `data.frame`

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  - **Indices:** e.g., the third column has index 3
  - **Logical:** A vector of TRUE and FALSE
  - **Names:** They must be in quotation marks

- Example:

```
fb[1, c("Venue", "Attendance")]
```

```
##             Venue Attendance
## 1 Stade Louis II.       7500
```

2. Anatomy of a `data.frame`

2.4. Subsetting

- But **$** is not the only way to **extract** a variable from a dataset
  - You can also make use of the **[ ]** subsetting operator

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  - **Indices:** e.g., the third column has index 3
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  - **Names:** They must be in quotation marks

- Example:

```
fb[1, c("Venue", "Attendance")]
```

```
##             Venue Attendance
## 1 Stade Louis II.       7500
```

- Brackets also work for vectors:

```
vec <- c(3, 2, 1)
vec[c(T, F, T)]
```

```
## [1] 3 1
```

Practice

1) Download and import the dataset if you haven't already

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2) Combine the use of `[ ]` and `nrow()` to obtain the last value of the `Wk` variable

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2) Combine the use of `[ ]` and `nrow()` to obtain the last value of the `Wk` variable

3) Subset the home team, the score, and the away team for matches that occured during the last week

Practice

05:59

1) Download and import the dataset if you haven't already

2) Combine the use of `[ ]` and `nrow()` to obtain the last value of the `Wk` variable

3) Subset the home team, the score, and the away team for matches that occured during the last week

You've got 6 minutes!

Solution

1) Download and import the dataset if you haven't already

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```
fb <- rio::import(here::here("lecture1", "ligue1.csv"))
```

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Solution

1) Download and import the dataset if you haven't already

```
fb <- rio::import(here::here("lecture1", "ligue1.csv"))
```

2) Combine the use of `[ ]` and `nrow()` to obtain the last value of the `Wk` variable

```
last_week <- fb[nrow(fb), "Wk"]
last_week
```

```
## [1] 38
```

Solution

3) Subset the home team, the score, and the away team for matches that occured during the last week

Solution

3) Subset the home team, the score, and the away team for matches that occured during the last week

```
fb[Wk == last_week, c("Home", "Score", "Away")]
```

```
## Error: object 'Wk' not found
```

Solution

3) Subset the home team, the score, and the away team for matches that occured during the last week

```
fb[Wk == last_week, c("Home", "Score", "Away")]
```

```
## Error: object 'Wk' not found
```

- Oops! Seems like **R couldn't find** the Wk variable
  - R was looking for Wk **in our environment**
  - But there is no Wk there

- We must **refer to fb** which is in our environment
  - Then we can **access Wk using the $** symbol

```
fb[fb$Wk == 38, c("Home", "Score", "Away")]
```

Solution

3) Subset the home team, the score, and the away team for matches that occured during the last week

```
fb[Wk == last_week, c("Home", "Score", "Away")]
```

```
## Error: object 'Wk' not found
```

- Oops! Seems like **R couldn't find** the Wk variable
  - R was looking for Wk **in our environment**
  - But there is no Wk there

- We must **refer to fb** which is in our environment
  - Then we can **access Wk using the $** symbol

```
fb[fb$Wk == 38, c("Home", "Score", "Away")]
```

```
##                Home Score          Away
## 371           Lille   2-2        Rennes
## 372           Brest   2-4      Bordeaux
## 373          Nantes   1-1 Saint-Étienne
## 374 Clermont Foot   1-2          Lyon
## 375          Angers   2-0   Montpellier
## 376         Lorient   1-1        Troyes
## 377      Paris S-G   5-0          Metz
## 378           Reims   2-3          Nice
## 379      Marseille   4-0    Strasbourg
## 380            Lens   2-2        Monaco
```

Overview

1. Getting started ✓
- 1.1. About R
- 1.2. The R Studio IDE
- 1.3. R projects
- 1.4. Packages
- 1.5. Import and eyeball data

2. Anatomy of a data.frame ✓
- 2.1. Data structure
- 2.2. Classes
- 2.3. Vectors
- 2.4. Subsetting

3. The dplyr package
- 3.1. Basic functions
- 3.2. group_by() and summarise()

4. A few words on learning R
- 4.1. When it doesn't work the way you want
- 4.2. Where to find help
- 4.3. When it doesn't work at all

5. Wrap up!

3. The `dplyr` package

3.1. Basic functions

`dplyr` is a **grammar** of data manipulation providing very **user-friendly functions** to handle the most common **data manipulation** tasks:

3. The `dplyr` package

3.1. Basic functions

`dplyr` is a **grammar** of data manipulation providing very **user-friendly functions** to handle the most common **data manipulation** tasks:

- `mutate()`: add/modify variables
- `select()`: keep/drop variables (columns)
- `filter()`: keep/drop observations (rows)
- `arrange()`: sort rows according to the values of given variable(s)
- `summarise()`: aggregate the data into descriptive statistics

3. The `dplyr` package

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3. The `dplyr` package

3.1. Basic functions

`dplyr` is a **grammar** of data manipulation providing very **user-friendly functions** to handle the most common **data manipulation** tasks:

- `mutate()`: add/modify variables
- `select()`: keep/drop variables (columns)
- `filter()`: keep/drop observations (rows)
- `arrange()`: sort rows according to the values of given variable(s)
- `summarise()`: aggregate the data into descriptive statistics

- A very handy **operator** to use with the **dplyr** grammar is the **pipe %>%**
  - You can basically read **a %>% b()** as *"apply function b() to object a"*
  - With this operator you can easily **chain the operations** you apply to an object

3. The `dplyr` package

3.1. Basic functions

```
fb
                                                                  #
                                                                  #
                                                                  #
                                                                  #
                                                                  #
                                                                  #
```

```
##      Wk Day       Date  Time          Home  xG Score xG.1          Away Attendance  ...
## 1     1 Fri 2021-08-06 21:00        Monaco 2.0   1-1  0.3        Nantes       7500  ...
## 2     1 Sat 2021-08-07 17:00          Lyon 1.4   1-1  0.8         Brest      29018  ...
## 3     1 Sat 2021-08-07 21:00        Troyes 0.8   1-2  1.2     Paris S-G      15248  ...
## 4     1 Sun 2021-08-08 13:00        Rennes 0.6   1-1  2.0          Lens      22567  ...
## 5     1 Sun 2021-08-08 15:00      Bordeaux 0.7   0-2  3.3 Clermont Foot      18748  ...
## 6     1 Sun 2021-08-08 15:00    Strasbourg 0.4   0-2  0.9        Angers      23250  ...
## 7     1 Sun 2021-08-08 15:00          Nice 0.8   0-0  0.2         Reims      18030  ...
## 8     1 Sun 2021-08-08 15:00 Saint-Étienne 2.1   1-1  1.3       Lorient      20461  ...
## 9     1 Sun 2021-08-08 17:00          Metz 0.7   3-3  1.4         Lille      15551  ...
##  ...   ... ...         ...   ...           ... ...   ...  ...           ...         ...  ...
```

3. The `dplyr` package

3.1. Basic functions

```
fb %>%
  select(Home, xG, Score, xG.1, Away)                # Keep/drop certain columns
                                                     #
                                                     #
                                                     #
                                                     #
                                                     #
```

```
##              Home  xG Score xG.1          Away
## 1          Monaco 2.0   1-1  0.3        Nantes
## 2            Lyon 1.4   1-1  0.8         Brest
## 3          Troyes 0.8   1-2  1.2     Paris S-G
## 4          Rennes 0.6   1-1  2.0          Lens
## 5        Bordeaux 0.7   0-2  3.3 Clermont Foot
## 6      Strasbourg 0.4   0-2  0.9        Angers
## 7            Nice 0.8   0-0  0.2         Reims
## 8   Saint-Étienne 2.1   1-1  1.3       Lorient
## 9            Metz 0.7   3-3  1.4         Lille
##  ...             ... ...   ...  ...           ...
```

3. The `dplyr` package

3.1. Basic functions

```
fb %>%
  select(Home, xG, Score, xG.1, Away) %>%          # Keep/drop certain columns
  mutate(home_winner = xG > xG.1)                  # Create a new variable
                                                   #
                                                   #
                                                   #
                                                   #
```

```
##              Home  xG Score xG.1          Away home_winner
## 1          Monaco 2.0   1-1  0.3        Nantes        TRUE
## 2            Lyon 1.4   1-1  0.8         Brest        TRUE
## 3          Troyes 0.8   1-2  1.2     Paris S-G       FALSE
## 4          Rennes 0.6   1-1  2.0          Lens       FALSE
## 5        Bordeaux 0.7   0-2  3.3 Clermont Foot       FALSE
## 6      Strasbourg 0.4   0-2  0.9        Angers       FALSE
## 7            Nice 0.8   0-0  0.2         Reims        TRUE
## 8   Saint-Étienne 2.1   1-1  1.3       Lorient        TRUE
## 9            Metz 0.7   3-3  1.4         Lille       FALSE
##  ...             ... ...   ...  ...           ...         ...
```

3. The `dplyr` package

3.1. Basic functions

```
fb %>%
  select(Home, xG, Score, xG.1, Away) %>%          # Keep/drop certain columns
  mutate(home_winner = xG > xG.1) %>%              # Create a new variable
  filter(Home == "Rennes")                         # Keep/drop certain rows
                                                   #
                                                   #
                                                   #
```

```
##      Home  xG Score xG.1          Away home_winner
## 1  Rennes 0.6   1-1  2.0          Lens       FALSE
## 2  Rennes 0.9   1-0  0.5        Nantes        TRUE
## 3  Rennes 1.0   0-2  0.5         Reims        TRUE
## 4  Rennes 2.4   6-0  0.3 Clermont Foot        TRUE
## 5  Rennes 0.8   2-0  1.4     Paris S-G       FALSE
## 6  Rennes 1.5   1-0  0.6    Strasbourg        TRUE
## 7  Rennes 3.8   4-1  1.1          Lyon        TRUE
## 8  Rennes 3.1   2-0  0.7   Montpellier        TRUE
## 9  Rennes 0.8   1-2  0.6         Lille        TRUE
         ... ...   ...  ...           ...         ...
```

3. The `dplyr` package

3.1. Basic functions

```
fb %>%
  select(Home, xG, Score, xG.1, Away) %>%          # Keep/drop certain columns
  mutate(home_winner = xG > xG.1) %>%              # Create a new variable
  filter(Home == "Rennes") %>%                     # Keep/drop certain rows
  arrange(-xG)                                     # Sort rows
                                                   #
                                                   #
```

```
##      Home  xG Score xG.1          Away home_winner
## 1  Rennes 3.8   4-1  1.1          Lyon        TRUE
## 2  Rennes 3.3   6-0  0.4      Bordeaux        TRUE
## 3  Rennes 3.3   6-1  0.9          Metz        TRUE
## 4  Rennes 3.1   2-0  0.7   Montpellier        TRUE
## 5  Rennes 2.7   2-0  0.3         Brest        TRUE
## 6  Rennes 2.6   4-1  0.4        Troyes        TRUE
## 7  Rennes 2.4   6-0  0.3 Clermont Foot        TRUE
## 8  Rennes 1.9   2-3  2.9        Monaco       FALSE
## 9  Rennes 1.7   2-0  0.3        Angers        TRUE
         ... ...   ...  ...           ...         ...
```

3. The `dplyr` package

3.1. Basic functions

```
fb %>%
  select(Home, xG, Score, xG.1, Away) %>%          # Keep/drop certain columns
  mutate(home_winner = xG > xG.1) %>%              # Create a new variable
  filter(Home == "Rennes") %>%                     # Keep/drop certain rows
  arrange(-xG) %>%                                 # Sort rows
  summarise(expected_wins = mean(home_winner),     # Aggregate into statistics
            expected_goals = sum(xG))              #
```

```
##   expected_wins expected_goals
## 1     0.8421053           36.6
```

3. The `dplyr` package

3.1. Basic functions

- Here are two very **handy functions** to use within `mutate()`

3. The `dplyr` package

3.1. Basic functions

- Here are two very **handy functions** to use within `mutate()`

ifelse

```
fb %>%
  select(Home, Attendance) %>%
  mutate(att_bin = ifelse(Attendance > 10000,
                          "Large",
                          "Low")
         ) %>% head()
```

```
##          Home Attendance att_bin
## 1      Monaco       7500     Low
## 2        Lyon      29018   Large
## 3      Troyes      15248   Large
## 4      Rennes      22567   Large
## 5   Bordeaux      18748   Large
## 6 Strasbourg      23250   Large
```

3. The `dplyr` package

3.1. Basic functions

- Here are two very **handy functions** to use within `mutate()`

ifelse

```
fb %>%
  select(Home, Attendance) %>%
  mutate(att_bin = ifelse(Attendance > 10000,
                          "Large",
                          "Low")
         ) %>% head()
```

```
##          Home Attendance att_bin
## 1      Monaco       7500     Low
## 2        Lyon      29018   Large
## 3      Troyes      15248   Large
## 4      Rennes      22567   Large
## 5   Bordeaux      18748   Large
## 6 Strasbourg      23250   Large
```

case_when

```
fb %>%
  select(Home, xG, xG.1, Away) %>%
  mutate(xWin = case_when(xG > xG.1 ~ "Home",
                          xG == xG.1 ~ "Draw",
                          xG < xG.1 ~ "Away")
         ) %>% head()
```

```
##          Home  xG xG.1          Away xWin
## 1      Monaco 2.0  0.3        Nantes Home
## 2        Lyon 1.4  0.8         Brest Home
## 3      Troyes 0.8  1.2     Paris S-G Away
## 4      Rennes 0.6  2.0          Lens Away
## 5   Bordeaux 0.7  3.3 Clermont Foot Away
## 6 Strasbourg 0.4  0.9        Angers Away
```

3. The `dplyr` grammar

3.3. group_by() and summarise()

- With `group_by()` you can perform **computations separately** for the different **categories of a variable**

3. The `dplyr` grammar

3.3. group_by() and summarise()

- With `group_by()` you can perform **computations separately** for the different **categories of a variable**

```
fb %>%
  select(Wk, Home, xG) %>%
  mutate(all.xG = mean(xG)) %>%
  head(10)
```

```
##    Wk          Home  xG   all.xG
## 1   1        Monaco 2.0 1.473421
## 2   1          Lyon 1.4 1.473421
## 3   1        Troyes 0.8 1.473421
## 4   1        Rennes 0.6 1.473421
## 5   1      Bordeaux 0.7 1.473421
## 6   1    Strasbourg 0.4 1.473421
## 7   1          Nice 0.8 1.473421
## 8   1 Saint-Étienne 2.1 1.473421
## 9   1          Metz 0.7 1.473421
## 10  1   Montpellier 0.5 1.473421
```

3. The `dplyr` grammar

3.3. group_by() and summarise()

- With `group_by()` you can perform **computations separately** for the different **categories of a variable**

```
fb %>%
  select(Wk, Home, xG) %>%
  mutate(all.xG = mean(xG)) %>%
  head(10)
```

```
##    Wk          Home  xG   all.xG
## 1   1        Monaco 2.0 1.473421
## 2   1          Lyon 1.4 1.473421
## 3   1        Troyes 0.8 1.473421
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## 9   1          Metz 0.7 1.473421
## 10  1   Montpellier 0.5 1.473421
```

```
fb %>%
  select(Wk, Home, xG) %>%
  group_by(Home) %>%
  mutate(home.xG = mean(xG)) %>%
  head(6)
```

```
## # A tibble: 6 × 4
## # Groups:   Home [6]
##      Wk Home          xG home.xG
##   <int> <chr>      <dbl>   <dbl>
## 1     1 Monaco       2      1.69
## 2     1 Lyon         1.4    2.07
## 3     1 Troyes       0.8    1.21
## 4     1 Rennes       0.6    1.93
## 5     1 Bordeaux     0.7    1.23
## 6     1 Strasbourg   0.4    1.73
```

3. The `dplyr` grammar

3.3. group_by() and summarise()

- It is particularly **useful with summarise()**
  - summarise keeps the grouping variable
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3. The `dplyr` grammar

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```
fb %>%
  group_by(Wk) %>%
  summarise(n = n(),
            tot_xG = sum(xG)+sum(xG.1),
            avg_WG = tot_xG/n) %>%
  head(4)
```

```
## # A tibble: 4 × 4
##      Wk     n tot_xG avg_WG
##   <int> <int>  <dbl>  <dbl>
## 1     1    10   23.4   2.34
## 2     2    10   26.6   2.66
## 3     3    10   25.7   2.57
## 4     4    10   30.4   3.04
```

3. The `dplyr` grammar

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##      Wk     n tot_xG avg_WG
##   <int> <int>  <dbl>  <dbl>
## 1     1    10   23.4   2.34
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mutate() $\neq$ summarise()

- **mutate()** takes an operation that converts:
  - **A vector into another vector**
- **summarise()** takes an operation that converts:
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Ungrouping

- **group_by()** applies to all subsequent operations
- To cancel its effect you must **ungroup()** the data

```
fb %>%
  group_by(Wk) %>%
  mutate(test = mean(xG)) %>%
  ungroup() %>%
  ...
```

Practice

1) Start from the `fb` dataset and keep only the variables `Home`, `Score` and `Away`

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2) Use the `separate()` function from `tidyr` to split the `Score` variable into `home_score` and `away_score`

```
data.frame(x = "a_b") %>%
  separate(x, c("x", "y"), "_")
```

```
##   x y
## 1 a b
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09:59

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You've got 10 minutes!

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```
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##     Home Score   Away
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## 2   Lyon   1-1  Brest
```

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```
fb %>%
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  head(2)
```

```
##     Home home_score away_score   Away
## 1 Monaco          1          1 Nantes
## 2   Lyon          1          1  Brest
```

Solution

3) Convert these two variables into numeric vectors

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  select(Home, Score, Away) %>%
  separate(Score, c("home_score", "away_score"), "-") %>%
  mutate(home_score = as.numeric(home_score),
         away_score = as.numeric(away_score),
         winner = case_when(home_score < away_score ~ "Away",
                            home_score == away_score ~ "Draw",
                            home_score > away_score ~ "Home")) %>%
  head()
```

Solution

3) Convert these two variables into numeric vectors

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```
fb %>%
  select(Home, Score, Away) %>%
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```

```
##         Home home_score away_score          Away winner
## 1     Monaco          1          1        Nantes   Draw
## 2       Lyon          1          1         Brest   Draw
## 3     Troyes          1          2     Paris S-G   Away
## 4     Rennes          1          1          Lens   Draw
## 5   Bordeaux          0          2 Clermont Foot   Away
## 6 Strasbourg          0          2        Angers   Away
```

Solution

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  group_by(winner) %>%
  summarise(pct = 100 * (n() / nrow(fb)))
```

Solution

5) Use `group_by()` and `summarise()` to compute the percentage of draws, home wins and away wins

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fb %>%
  select(Home, Score, Away) %>%
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```

```
## # A tibble: 3 × 2
##   winner   pct
##   <chr>  <dbl>
## 1 Away    30.5
## 2 Draw    26.8
## 3 Home    42.6
```

Overview

1. Getting started ✓
- 1.1. About R
- 1.2. The R Studio IDE
- 1.3. R projects
- 1.4. Packages
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2. Anatomy of a data.frame ✓
- 2.1. Data structure
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3. The dplyr package ✓
- 3.1. Basic functions
- 3.2. group_by() and summarise()

4. A few words on learning R
- 4.1. When it doesn't work the way you want
- 4.2. Where to find help
- 4.3. When it doesn't work at all

5. Wrap up!

4. A few words on learning R

4.1. When it doesn't work the way you want

- When things do not work the way you want, **NAs are the usual suspects**
  - For instance, this is how the mean function reacts to NAs:

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```
mean(c(1, 2, NA))
```

```
## [1] NA
```

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```
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```

```
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```

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mean(c(1, 2, NA), na.rm = T)
```

```
## [1] 1.5
```

- You should systematically **check for NAs!**

```
is.na(c(1, 2, NA))
```

```
## [1] FALSE FALSE  TRUE
```

4. A few words on learning R

4.1. When it doesn't work the way you want

- **Don't pipe blindfolded!**
  - **Check** that each command does what it's expected to do
  - View or print your data **at each step**

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```

```
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```

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- Oftentimes things don't work either because:
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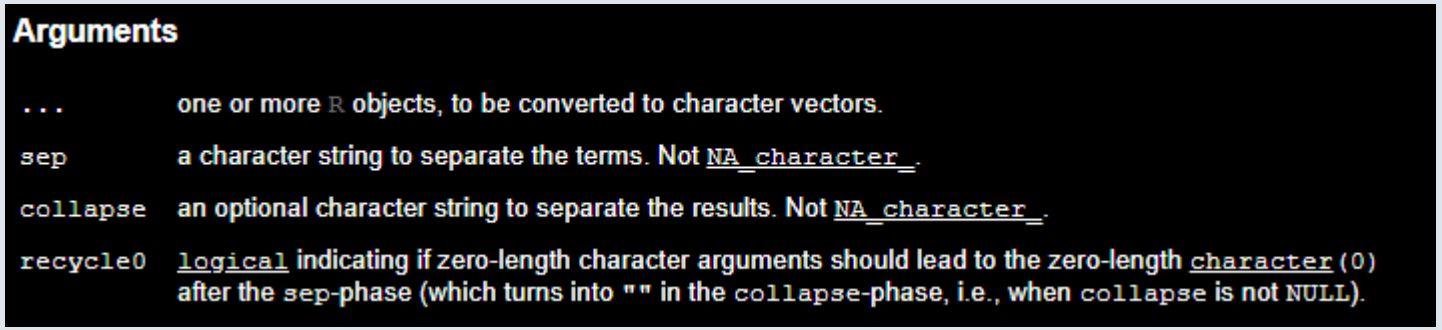

Arguments

`...`	one or more R objects, to be converted to character vectors.
`sep`	a character string to separate the terms. Not `NA_character_`.
`collapse`	an optional character string to separate the results. Not `NA_character_`.
`recycle0`	`logical` indicating if zero-length character arguments should lead to the zero-length `character(0)` after the `sep`-phase (which turns into `""` in the `collapse`-phase, i.e., when `collapse` is not `NULL`).

4. A few words on learning R

4.3. When it doesn't work at all

- Sometimes R breaks and returns an **error** (usually kind of cryptic)

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1. Look for **keywords** that might help you understand where it comes from
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4. A few words on learning R

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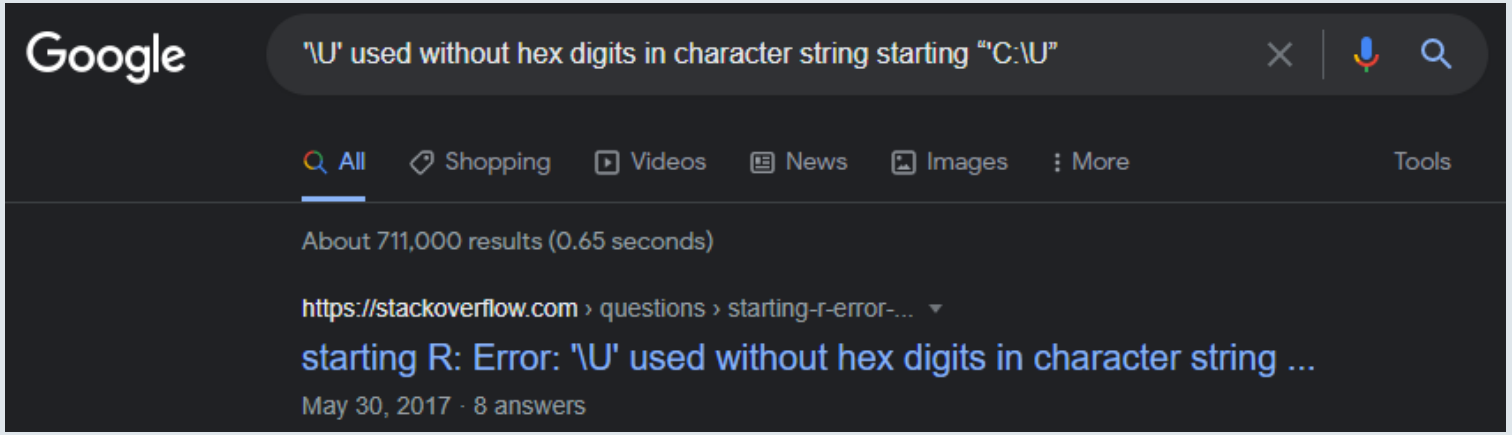

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5. Wrap up!

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1. Import data

```
fb <- rio::import(here::here("lecture1", "ligue1.csv"))
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is.numeric("1.6180339") # What would be the output?
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5. Wrap up!

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```

```
## [1] FALSE
```

3. Subsetting

```
fb$Home[3]
```

```
## [1] "Troyes"
```

5. Wrap up!

4. Packages

```
library(dplyr)
```

5. Wrap up!

4. Packages

```
library(dplyr)
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5. The dplyr grammar

Function	Meaning
mutate()	Modify or create a variable
select()	Keep a subset of variables
filter()	Keep a subset of observations
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